

Program Version: 5.3.1

Product: Oxygen Optode 4831UW DW

Serial No: 1300

Visual and Mechanical Checks:

- 1.1 Soldering quality
- 1.2 Visual surface
- 1.3 Galvanic isolation between housing and electronics

Current Drain and Voltages:

2.1	Average current drain at 0.5 Hz sampling (Max.: 33 mA)	23.5	mA
2.2	CANBus Current drain at 0.5 Hz sampling (Max.: 33 mA)		mA
2.3	Current drain in sleep (Max.: 270 μ A)	231	μ A
2.4	CANBus Current drain in sleep (Max.: 180 μ A)		μ A
2.5	DSP IO voltage, J4.18 (3.3 $\pm$ 0.15V)	3.32	V
2.6	DSP Core voltage, J4.17(1.8 $\pm$ 0.05 V)	1.81	V
2.7	Excitation driver voltage, C4 Analog Board (4.3 $\pm$ 0.1 V)	4.28	V

Performance test:

	Channel:	Blue	Red
3.1	Average of Receiver readings (0 $\pm$ 150mV)	-7.7 mV	-4.3 mV
3.2	Standard Deviation of Receiver readings (Max.: 45mV/10mV)	1.12 mV	0.18 mV
3.3	Amplitude measurement with non-fluorescence foil (<60mV/650-1200mV)	8.7 mV	1062.6 mV
3.4	CANBus Output test		

Function test from 0 to 40°C:

	Channel:	Blue	Red
4.1	Minimum amplitude measurement (Blue: >550 mV, Red >550 mV)	642.4 mV	828.3 mV
4.2	Maximum amplitude measurement (Blue: <1600 mV, Red <1400 mV)	953.4 mV	1132.6 mV
4.3	Minimum phase measurement (Blue: >32°, Red: >3°)	38.42 °	8.67 °
4.4	Maximum phase measurement (Blue: <45°, Red: <10°)	44.51 °	9.39 °
4.5	Maximum standard deviation of Phase measurement: (< 0.07°)	0.06 °	0.05 °
4.6	Minimum temperature raw data measurement: (<-200 mV)		-400.2 mV
4.7	Maximum temperature raw data measurement: (>450 mV)		745.8 mV

Date: 06 Oct 2025

Sign:

Andrine Dale Raknes

Andrine Raknes, Production Engineer

Program Version: 5.3.1

Product: Oxygen Optode 4831UW DW

Serial No: 1301

Visual and Mechanical Checks:

- 1.1 Soldering quality
- 1.2 Visual surface
- 1.3 Galvanic isolation between housing and electronics

Current Drain and Voltages:

2.1	Average current drain at 0.5 Hz sampling (Max.: 33 mA)	23.4	mA
2.2	CANBus Current drain at 0.5 Hz sampling (Max.: 33 mA)		mA
2.3	Current drain in sleep (Max.: 270 μ A)	231	μ A
2.4	CANBus Current drain in sleep (Max.: 180 μ A)		μ A
2.5	DSP IO voltage, J4.18 (3.3 ± 0.15 V)	3.32	V
2.6	DSP Core voltage, J4.17(1.8 ± 0.05 V)	1.82	V
2.7	Excitation driver voltage, C4 Analog Board (4.3 ± 0.1 V)	4.29	V

Performance test:

	Channel:	Blue	Red
3.1	Average of Receiver readings (0 ± 150 mV)	2.6 mV	0.6 mV
3.2	Standard Deviation of Receiver readings (Max.: 45mV/10mV)	2.51 mV	1.33 mV
3.3	Amplitude measurement with non-fluorescence foil (<60mV/650-1200mV)	10.2 mV	712.2 mV
3.4	CANBus Output test		

Function test from 0 to 40°C:

	Channel:	Blue	Red
4.1	Minimum amplitude measurement (Blue: >550 mV, Red >550 mV)	760.3 mV	830.1 mV
4.2	Maximum amplitude measurement (Blue: <1600 mV, Red <1400 mV)	1137.5 mV	1119 mV
4.3	Minimum phase measurement (Blue: >32°, Red: >3°)	37.07 °	8.14 °
4.4	Maximum phase measurement (Blue: <45°, Red: <10°)	42.9 °	8.9 °
4.5	Maximum standard deviation of Phase measurement: (< 0.07°)	0.05 °	0.05 °
4.6	Minimum temperature raw data measurement: (<-200 mV)		-472.1 mV
4.7	Maximum temperature raw data measurement: (>450 mV)		668.1 mV

Date: 06 Oct 2025

Sign:

Andrine Dale Raknes

Andrine Raknes, Production Engineer